

```
#!/bin/bash
# The 'Shebang' (#!/) tells the OS to use the Bash interpreter.
# 'echo' prints text to the console.

echo "Welcome to your first Linux Script!"
echo "Today is: $(date)"
```

```
#!/bin/bash
# Note: Do NOT put spaces around the '=' sign when assigning
variables.
num1=15
num2=25

# $(( )) is the syntax for integer arithmetic in Bash.
sum=$((num1 + num2))

echo "Calculating: $num1 + $num2"
echo "Result: $sum"
```

```
#!/bin/bash
# 'read' pauses execution and waits for user typing.
echo "What is your server's name?"
read server_name

echo "Configuring server: $server_name..."
sleep 1 # Pauses for 1 second for effect
echo "Server $server_name is now ONLINE."
```

```
#!/bin/bash
echo "Enter system access level (1-100):"
read level

# -ge means 'Greater than or Equal to'
if [ $level -ge 50 ]; then
    echo "Access GRANTED. Welcome, Admin."
else
    echo "Access DENIED. Minimum level 50 required."
fi
```

```
#!/bin/bash
folder="logs_backup"

# -d checks if a directory exists on the disk.
if [ -d "$folder" ]; then
    echo "Folder '$folder' already exists. No action taken."
else
    echo "Creating missing folder: $folder"
    mkdir $folder
fi
```

```
#!/bin/bash
# {1..5} creates a sequence from 1 to 5.
for i in $(seq 1 5)
do
    echo "Generating report_${i}.txt..."
    touch "report_${i}.txt"
done
echo "Batch creation complete."
```

```
#!/bin/bash
key="linux123"
guess=""

# Keep looping as long as the guess is NOT EQUAL (!=) to the key.
while [ "$guess" != "$key" ]
do
    echo "Enter System Key to unlock:"
    read guess
done

echo "UNLOCKED successfully."
```

```
#!/bin/bash
echo "Choose a Task: 1) View Disk 2) View Memory 3) View Users"
read task

case $task in
    1) df -h ;;          # Display disk usage
    2) free -m ;;        # Display memory usage
    3) whoami ;;         # Display current user
    *) echo "Invalid choice. Please pick 1, 2, or 3." ;;
esac
```

```
#!/bin/bash
# Define a reusable function.
health_check() {
    echo "--- SYSTEM HEALTH ---"
    uptime
    echo "--- ACTIVE CONNECTIONS ---"
    netstat -ant | wc -l
}

# Call the function.
health_check
```

```
#!/bin/bash
# 1. Variables
SOURCE_DIR="./projects"
BACKUP_DIR="./backups"
TIMESTAMP=$(date +%Y-%m-%d_%H%M)

# 2. Preparation
mkdir -p $BACKUP_DIR
echo "Starting backup of $SOURCE_DIR..."

# 3. Compression (tar -c:create, -z:zip, -v:verbose, -f:file)
tar -czvf $BACKUP_DIR/backup_$TIMESTAMP.tar.gz $SOURCE_DIR

# 4. Result
echo "Backup saved as backup_$TIMESTAMP.tar.gz in $BACKUP_DIR"
```